

Measurement, Units, Dimensions and Vectors, **solved**

Step-by-step solutions to every practice problem in Chapter 1. Try each problem on your own before you read its solution.

By **Pratik Bhanuse**, PhD Physics candidate, Western University ·
pratikbhanuse.github.io

P1.1

Find the dimensions of each quantity from its defining equation.

(a) Work: $W = Fd$, so $[W] = [MLT^{-2}][L] = [ML^2T^{-2}]$. Torque: $\tau = rF \sin \theta$, so $[\tau] = [L][MLT^{-2}] = [ML^2T^{-2}]$. **Same dimensions.** They are still different physical quantities: work is a scalar, while torque is a vector whose lever arm is perpendicular to the force.

(b) Impulse: $J = F\Delta t$, so $[J] = [MLT^{-2}][T] = [MLT^{-1}]$. Momentum: $p = mv$, so $[p] = [M][LT^{-1}] = [MLT^{-1}]$. **Same.** This is expected, because the impulse-momentum theorem says $J = \Delta p$.

(c) Pressure: F/A gives $[MLT^{-2}]/[L^2] = [ML^{-1}T^{-2}]$. Energy density: E/V gives $[ML^2T^{-2}]/[L^3] = [ML^{-1}T^{-2}]$. **Same.**

(d) Angular momentum: $L = mvr$, so $[L] = [M][LT^{-1}][L] = [ML^2T^{-1}]$. Planck's constant: $h = E/\nu$, so $[h] = [ML^2T^{-2}]/[T^{-1}] = [ML^2T^{-1}]$. **Same.** This is why Bohr could quantise angular momentum in units of $h/2\pi$ (Chapter 24).

All four pairs have the same dimensions.

P1.2

Two vectors. For $\vec{A} + \vec{B} = \vec{0}$ we need $\vec{B} = -\vec{A}$, which forces $|\vec{B}| = |\vec{A}|$. So two vectors of unequal magnitude can **never** add to zero. The smallest possible resultant is $|\vec{A} - \vec{B}| > 0$.

Three vectors. Yes, this is possible. If $\vec{A} + \vec{B} + \vec{C} = \vec{0}$, the three vectors placed head to tail form a closed triangle. A triangle can have three unequal sides, provided each side is no longer than the sum of the other two. Example: magnitudes 3, 4 and 5 arranged as a right triangle.

P1.3

(a) **0.004 5670**: the leading zeros are not significant. The digits 4, 5, 6, 7 and the trailing 0 after the decimal point are significant, giving 5 s.f. Rounded to 3 s.f.: **0.004 57**.

(b) **2.0063**: every digit is significant, including the zeros between non-zero digits. That is 5 s.f. The fourth digit is 6, which is greater than 5, so we round up: **2.01**.

(c) **6.0225×10^{23}** : 5 s.f. Rounded: **6.02×10^{23}** .

P1.4

$\vec{A} \cdot \vec{B} = AB \cos \theta = 0$ with $A, B \neq 0$ requires $\cos \theta = 0$, so $\theta = 90^\circ$.

$|\vec{A} \times \vec{B}| = AB \sin \theta = 0$ requires $\sin \theta = 0$, so $\theta = 0^\circ$ or 180° .

No angle satisfies both conditions. So it is **impossible** for two non-zero vectors. If both products vanish, at least one of the vectors must be the zero vector.

P1.5

Identify. By homogeneity, quantities that are added together must have the same dimensions.

Since a/V^2 is added to P , it must have the dimensions of pressure:

$$\left[\frac{a}{V^2} \right] = [P] = [ML^{-1}T^{-2}] \Rightarrow [a] = [ML^{-1}T^{-2}][L^3]^2 = [ML^5T^{-2}].$$

Since b is subtracted from V , it must be a volume:

$$[b] = [L^3].$$

Physically, b is the volume excluded by the gas molecules themselves, and a measures the strength of the attractions between them.

P1.6

Set up. Assume $v = kF^a\mu^b$, with k dimensionless. The dimensions are $[F] = [MLT^{-2}]$ and $[\mu] = [ML^{-1}]$.

Execute.

$$[LT^{-1}] = [MLT^{-2}]^a [ML^{-1}]^b = [M^{a+b} L^{a-b} T^{-2a}].$$

Comparing powers:

$$\begin{aligned} T : \quad -2a &= -1 \Rightarrow a = \frac{1}{2}, \\ M : \quad a + b &= 0 \Rightarrow b = -\frac{1}{2}, \\ L : \quad a - b &= \frac{1}{2} + \frac{1}{2} = 1. \checkmark \end{aligned}$$

So

$$v = k \sqrt{\frac{F}{\mu}}.$$

A full treatment (Chapter 11) gives $k = 1$.

Evaluate. A tighter string (larger F) carries waves faster, and a heavier string (larger μ) carries them more slowly. Both agree with experience.

P1.7

Set up. Take east as $+x$ and north as $+y$, and write each displacement in components.

- $\vec{d}_1 = 30 \hat{j} \text{ m}$
- $\vec{d}_2 = 20 \hat{i} \text{ m}$
- $\vec{d}_3$: $30\sqrt{2} \text{ m}$ towards the south-west, which is 45° between $-x$ and $-y$:

$$\vec{d}_3 = -30\sqrt{2} \cos 45^\circ \hat{i} - 30\sqrt{2} \sin 45^\circ \hat{j} = -30 \hat{i} - 30 \hat{j} \text{ m}.$$

Execute.

$$\vec{D} = (0 + 20 - 30) \hat{i} + (30 + 0 - 30) \hat{j} = -10 \hat{i} \text{ m}.$$

Result: the displacement is 10 m due west.

P1.8

The scalar component of $\vec{A}$ along $\vec{B}$ is $\vec{A} \cdot \hat{B}$, where $\hat{B} = \vec{B}/B$.

$$\hat{B} = \frac{\hat{i} + \hat{j}}{\sqrt{2}}, \quad \vec{A} \cdot \hat{B} = \frac{(3)(1) + (4)(1)}{\sqrt{2}} = \frac{7}{\sqrt{2}} \approx 4.95.$$

The vector component is the scalar component times the unit vector:

$$(\vec{A} \cdot \hat{B}) \hat{B} = \frac{7}{\sqrt{2}} \cdot \frac{\hat{i} + \hat{j}}{\sqrt{2}} = \frac{7}{2}(\hat{i} + \hat{j}).$$

Check. The part of $\vec{A}$ left over, $\vec{A} - \frac{7}{2}(\hat{i} + \hat{j}) = -\frac{1}{2}\hat{i} + \frac{1}{2}\hat{j}$, should be perpendicular to $\vec{B}$:
 $(-\frac{1}{2})(1) + (\frac{1}{2})(1) = 0$. ✓

P1.9

Central value: $R = 100/10 = 10 \Omega$.

For a quotient, the relative errors add:

$$\frac{\Delta R}{R} = \frac{\Delta V}{V} + \frac{\Delta I}{I} = \frac{5}{100} + \frac{0.2}{10} = 0.05 + 0.02 = 0.07 = 7\%.$$

Absolute error: $\Delta R = 0.07 \times 10 \Omega = 0.7 \Omega$.

$$R = (10.0 \pm 0.7) \Omega \quad (7\%).$$

P1.10

Angle. Use $R^2 = A^2 + B^2 + 2AB \cos \theta$:

$$169 = 25 + 144 + 2(5)(12) \cos \theta \Rightarrow 120 \cos \theta = 0 \Rightarrow \theta = 90^\circ.$$

This makes sense: $5^2 + 12^2 = 13^2$, a Pythagorean triple, so the vectors must be perpendicular.

Difference.

$$|\vec{A} - \vec{B}| = \sqrt{A^2 + B^2 - 2AB \cos \theta} = \sqrt{25 + 144 - 0} = 13 \text{ units.}$$

For perpendicular vectors, the sum and the difference always have equal magnitudes: they are the two diagonals of a rectangle.

P1.11

Volume of the room: $V = 10 \times 8 \times 3 = 240 \text{ m}^3$.

Mass of air:

$$m = \rho V = (1.2 \text{ kg/m}^3)(240 \text{ m}^3) \approx 290 \text{ kg} \approx 3 \times 10^2 \text{ kg}.$$

A typical student has a mass of about 60–70 kg, so the air in the room weighs about as much as **four or five students**. Most people guess far less.

P1.12

Identify. This is a chain of unit conversions: dose rate ($\mu\text{g}/\text{min}$) $\rightarrow$ mass rate (mg/h) $\rightarrow$ volume rate (mL/h).

Step 1. Dose per minute.

$$5.0 \frac{\mu\text{g}}{\text{kg min}} \times 70 \text{ kg} = 350 \mu\text{g}/\text{min}.$$

Step 2. Convert to mg/h.

$$350 \frac{\mu\text{g}}{\text{min}} \times \frac{1 \text{ mg}}{1000 \mu\text{g}} \times \frac{60 \text{ min}}{1 \text{ h}} = 21 \text{ mg}/\text{h}.$$

Step 3. Concentration of the bag.

$$c = \frac{400 \text{ mg}}{250 \text{ mL}} = 1.6 \text{ mg}/\text{mL}.$$

Step 4. Pump rate.

$$\text{rate} = \frac{21 \text{ mg}/\text{h}}{1.6 \text{ mg}/\text{mL}} = 13.1 \text{ mL}/\text{h} \approx 13 \text{ mL}/\text{h}.$$

The error. Entering 13.1 as mL/min delivers $13.1 \times 60 = 788 \text{ mL}/\text{h}$. That is an overdose by a factor of 60, which could be fatal for a drug that acts on the heart. Smart infusion pumps hold drug libraries with dose limits for exactly this reason.

P1.13

Idea. The *physical* energy is the same in both systems. Only the number and the unit change, so $n_1 u_1 = n_2 u_2$.

Energy has dimensions $[ML^2T^{-2}]$, so the new unit of energy is

$$u_2 = (\alpha \text{ kg})(\beta \text{ m})^2(\gamma \text{ s})^{-2} = \frac{\alpha\beta^2}{\gamma^2} \text{ J}.$$

Then

$$n_2 = n_1 \frac{u_1}{u_2} = 1 \times \frac{1 \text{ J}}{(\alpha\beta^2/\gamma^2) \text{ J}} = \frac{\gamma^2}{\alpha\beta^2}.$$

So $1 \text{ J} = \gamma^2/(\alpha\beta^2)$ new units of energy. This agrees with the general formula

$n_2 = n_1(M_1/M_2)^1(L_1/L_2)^2(T_1/T_2)^{-2}$ with $M_1/M_2 = 1/\alpha$, $L_1/L_2 = 1/\beta$ and $T_1/T_2 = 1/\gamma$.

Check. If the new mass unit is larger ($\alpha > 1$), 1 J should be *fewer* new units. The formula agrees. ✓

P1.14

Idea. The area of a triangle is half the area of the parallelogram spanned by two of its sides:

$$\frac{1}{2} |\vec{PQ} \times \vec{PR}|.$$

$$\vec{PQ} = Q - P = (-1, 2, 0), \quad \vec{PR} = R - P = (-1, 0, 3).$$

$$\vec{PQ} \times \vec{PR} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -1 & 2 & 0 \\ -1 & 0 & 3 \end{vmatrix} = \hat{i}(6 - 0) - \hat{j}(-3 - 0) + \hat{k}(0 + 2) = 6\hat{i} + 3\hat{j} + 2\hat{k}.$$

$$|\vec{PQ} \times \vec{PR}| = \sqrt{36 + 9 + 4} = 7.$$

$$\text{Area} = \frac{1}{2}(7) = 3.5 \text{ m}^2.$$

$$\text{Unit normal: } \hat{n} = \frac{6\hat{i} + 3\hat{j} + 2\hat{k}}{7}, \text{ or its negative.}$$

Check. The plane through the three intercepts is $x + y/2 + z/3 = 1$. Its normal is $(1, \frac{1}{2}, \frac{1}{3})$, which is proportional to $(6, 3, 2)$. ✓

P1.15

(a) Separate the variables in $\frac{dx}{dt} = \beta\sqrt{x}$:

$$\int_0^x x^{-1/2} dx = \int_0^t \beta dt \Rightarrow 2\sqrt{x} = \beta t \Rightarrow x = \frac{\beta^2 t^2}{4}.$$

(b) Differentiate:

$$v = \frac{dx}{dt} = \frac{\beta^2 t}{2}, \quad a = \frac{dv}{dt} = \frac{\beta^2}{2} \quad (\text{constant}).$$

Alternatively, using $a = v dv/dx$:

$$a = \beta\sqrt{x} \cdot \frac{\beta}{2\sqrt{x}} = \frac{\beta^2}{2}. \checkmark$$

(c) The time to cover distance s follows from $2\sqrt{s} = \beta t_s$, so $t_s = 2\sqrt{s}/\beta$. Then

$$\bar{v} = \frac{s}{t_s} = \frac{s\beta}{2\sqrt{s}} = \frac{\beta\sqrt{s}}{2}.$$

Check. The acceleration is constant and the particle starts from rest, so the average velocity should be half the final velocity. The final velocity is $v(s) = \beta\sqrt{s}$, and half of that is $\beta\sqrt{s}/2$. $\checkmark$

P1.16

Set up. Assume $f = k r^a \rho^b S^c$ with k dimensionless. The dimensions are:

- $[f] = [T^{-1}]$
- $[r] = [L]$
- $[\rho] = [ML^{-3}]$
- $[S] = [\text{force/length}] = [MLT^{-2}]/[L] = [MT^{-2}]$

Execute.

$$[T^{-1}] = [L]^a [ML^{-3}]^b [MT^{-2}]^c = [M^{b+c} L^{a-3b} T^{-2c}].$$

$$T: \quad -2c = -1 \Rightarrow c = \frac{1}{2},$$

$$M: \quad b + c = 0 \Rightarrow b = -\frac{1}{2},$$

$$L: \quad a - 3b = 0 \Rightarrow a = -\frac{3}{2}.$$

So

$$f = k\sqrt{\frac{S}{\rho r^3}}.$$

Doubling the radius. $f \propto r^{-3/2}$, so

$$\frac{f_2}{f_1} = 2^{-3/2} = \frac{1}{2\sqrt{2}} \approx 0.35.$$

The frequency falls to about 35% of its original value. Larger drops wobble more slowly, as you might expect.

P1.17

(a) Write each factor in base units:

- $[\rho] = \text{kg m}^{-3}$
- $[v] = \text{m s}^{-1}$
- $[D] = \text{m}$
- $[\eta] = \text{Pa s} = \text{N m}^{-2} \text{s} = (\text{kg m s}^{-2}) \text{m}^{-2} \text{s} = \text{kg m}^{-1} \text{s}^{-1}$

Then

$$[Re] = \frac{(\text{kg m}^{-3})(\text{m s}^{-1})(\text{m})}{\text{kg m}^{-1} \text{s}^{-1}} = \frac{\text{kg m}^{-1} \text{s}^{-1}}{\text{kg m}^{-1} \text{s}^{-1}} = 1.$$

So Re is dimensionless. ✓

(b) At rest:

$$Re = \frac{(1060)(0.30)(0.025)}{3.5 \times 10^{-3}} = \frac{7.95}{3.5 \times 10^{-3}} \approx 2.3 \times 10^3.$$

During exercise the speed is four times larger, so $Re \approx 9.1 \times 10^3$.

Comment. At rest, the mean flow sits near the laminar-turbulent transition. In fact flow in the aorta is pulsatile, and peak speeds during each heartbeat are higher. During exercise Re is well into the turbulent range. Turbulence in large arteries is one source of the sounds a doctor hears through a stethoscope. Abnormal turbulence, for example downstream of a narrowed heart valve, produces a *murmur*.

(c) Set $Re = 2000$ and solve for v :

$$v = \frac{Re \eta}{\rho D} = \frac{2000 \times 1.0 \times 10^{-3}}{1000 \times 0.020} = 0.10 \text{ m/s.}$$

Household water typically flows at 1–2 m/s, so flow in building pipes is almost always **turbulent**. Engineers therefore use turbulent-flow friction factors when they size pipes and pumps.

All worked examples and practice problems in these notes are original.

Textbooks for further reading and practice, with a chapter-by-chapter map, are listed in the References.

© 2026 Pratik Bhanuse. All rights reserved. · pratikbhanuse.github.io